

A Relational Observable Substrate and its NCG Realization

The Universal Calculus, the Connes–Chern Character,
and the Disclinated Haldane Class

Working draft — two-stage formulation

local observable algebras and strict descent first; the universal differential calculus as the (free, canonical) differential
and the cyclic characteristic map as the invariant selector; the crystalline equivalence as foundational hypothesis

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Abstract

The relational origin is a local observable net $\mathcal{A} : \text{Reg} \rightarrow C^*\text{Alg}$, $O \mapsto \mathcal{A}(O)$, whose Stage-1 fibre over a patch U_i is a full local observable algebra $\mathcal{A}_i = \mathcal{A}(U_i)$ — not a pre-selected occupied subspace. The occupied subspace, the Fermi projection, states, measurement effects and local unitaries are all selected inside $\mathcal{A}_i$ only after a realization or readout has been chosen. The relational layer is built as strict descent data: to each patch one attaches a local algebra-sheaf $\mathcal{A}_i$ on U_i , glued by comparison $*$ -isomorphisms on the overlaps, $\alpha_{ij} : \mathcal{A}_i|_{U_{ij}} \xrightarrow{\sim} \mathcal{A}_j|_{U_{ij}}$, satisfying the strict cocycle $\alpha_{ij}\alpha_{jk} = \alpha_{ik}$ on triple overlaps. The relational sheaf $\mathcal{F}_{\text{rel}}(U_i) = \mathcal{A}_i$, glued by the α_{ij} , is genuinely relational — and flat — before it is spectral; the groupoid $\mathcal{R}_Q \rightrightarrows I$ survives only as a presentation of the comparison site $K = N(\mathcal{R}_Q)$, no longer foundational. Noncommutative geometry enters at exactly one place: a realization functor $\pi_{\text{NCG}} : \mathcal{R}_Q \rightarrow \text{Hilb}_{\mathcal{A}}$ that represents the local algebras and implements the comparisons,

$$\mathcal{A}_i \mapsto (\pi_i : \mathcal{A}_i \rightarrow B(H_i)), \quad \alpha_{ij} \mapsto V_{ij}, \quad V_{ij}\pi_i(a)V_{ij}^* = \pi_j(\alpha_{ij}(a)),$$

the V_{ij} being functorial implementers of the strict descent isomorphisms α_{ij} , not the Berry/Kato parallel transports of the occupied bundle; the Wilson/Kato links of the physical occupied subspace appear only downstream of sector selection, and their holonomy is the realized defect class, not a datum of the descent cocycle. The differential is the *universal* calculus $\Omega_u^\bullet(\mathcal{A}_G)$ — canonical, functorial, and carried along by the substrate, so the α_{ij} lift to it for free and it stays strictly flat. The realization’s essential analytic datum is therefore not a Dirac operator but a faithful trace T on $\mathcal{A}_G$ for the integration; the noncommutative content that distinguishes invariants is the cyclic characteristic map. Because π_{NCG} is a functor the comparisons stay strictly composable, $V_{ij}V_{jk} = V_{ik}$, so the substrate carries no curvature; curvature is the universal Grassmann/Bellissard form, born the first time d acts on the selected projection. The connection is $a_{ij} = dV_{ij}$ and the curvature is $F = P(dP)^2$, simplicially $F_{ijk}^{\text{rel}} = P_i dV_{ij} dV_{jk}$. Closure is the Bianchi identity, automatic from $P^2 = P$ and the Leibniz rule — and it is strict composability of the implementers that protects it; no twisted algebra and no associativity theorem are needed. A characteristic map — a cyclic cocycle τ , here the Connes–Chern/Bellissard area cocycle built from the trace T and the position derivations $\partial_j = [X_j, \cdot]$ — pairs the universal curvature into a scalar Čech–Connes cochain, with class $[\check{c}] \in \check{H}^2(N(\mathcal{R}_Q); \mathbb{C})$. The bulk absolute number integralizes as the Connes pairing $\langle \tau, [P] \rangle \in \mathbb{Z}$ (an index), while the defect relative class integralizes through the determinant transition functions and the exponential sequence on the disk; in both forms one obtains $c_1^{\text{rel}} \in \check{H}^2(N(\mathcal{R}_Q); \mathbb{Z})$. Swapping τ targets higher Chern, $\mathbb{Z}_2$, gap-labeling, or Floquet invariants on the same substrate, and a non-index cocycle yields non-quantized spectral data. Under a deferred crystalline comparison hypothesis c_1^{rel} recovers the Bloch–Berry Chern class of the periodic Haldane model. We localize c_1^{rel} to the star $\text{St}(d)$ of a pentagonal disclination and give a finite to-do list; notably the defect decision reduces to transport determinants and so does not require the characteristic map’s derivations. Two constructions are left honestly open: the position 1-cocycle defining the characteristic map (which carries the invariant-selecting geometry) and the realization-functor comparison.

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Architecture. Two stages; the differential is universal and the characteristic map selects the invariant:

$$\begin{array}{c}
 \mathcal{A} : \text{Reg} \rightarrow C^* \text{Alg}, \quad \mathcal{F}_{\text{rel}}(U_i) = \mathcal{A}_i, \quad \alpha_{ij}\alpha_{jk} = \alpha_{ik} \\
 \underbrace{\hspace{15em}} \\
 \text{Stage 1 + 1.5: local algebras, strict descent, flat} \\
 \\
 \xrightarrow{\pi_{\text{NCG}}, \Omega_u} \underbrace{a_{ij} = dV_{ij}, \quad F = P(dP)^2, \quad \text{Bianchi free}}_{\text{Stage 2: represent + trace } T, \text{ universal } d} \xrightarrow{\chi_\tau} c_1^{\text{rel}} \in \check{H}^2(N(\mathcal{R}_Q); \mathbb{Z})
 \end{array}$$

1 Stage 1: the local observable substrate

The substrate records what the local quantum world is made of, using observable algebras rather than a pre-selected sector. It is built in two strictly flat layers: a local observable net that assigns an algebra to each region (the carrier, §1.1), and a strict relational descent layer that glues these into a sheaf of observable algebras over a comparison site (Stage 1.5, §1.2). Both layers are curvature-free by design; the comparisons compose on the nose, so the substrate is flat before it is spectral.

1.1 The local observable net (the carrier)

Let Reg be the poset of admissible regions/patches, ordered by inclusion. The carrier is a local observable net — a functor

$$\mathcal{A} : \text{Reg} \longrightarrow C^* \text{Alg}, \quad O \longmapsto \mathcal{A}(O), \quad (1)$$

isotonic: an inclusion $O \subseteq O'$ induces a unital embedding $\iota_{OO'} : \mathcal{A}(O) \hookrightarrow \mathcal{A}(O')$, functorially in inclusions. We record the variance of these maps explicitly, since it governs the descent construction of §1.2.

Lemma 1.1 (Variance of the carrier). *The assignment $O \mapsto \mathcal{A}(O)$ with the embeddings $\iota_{OO'}$ is a covariant functor $(\text{Reg}, \subseteq) \rightarrow C^*\text{Alg}$; equivalently a precosheaf of C^* -algebras on the cover, its gluing being the inductive (colimit) extension to the quasilocal algebra $\varinjlim_O \mathcal{A}(O)$. Consequently it is not a presheaf: isotony supplies co-restriction (extension) maps from smaller to larger regions, whereas a sheaf of observable algebras requires restriction maps from larger to smaller. Restriction maps are therefore not furnished by the net; they are an additional datum, a chosen localization $\rho_{O'O} : \mathcal{A}(O') \rightarrow \mathcal{A}(O)$ for $O \subseteq O'$ (e.g. a normal conditional expectation, a germ/dual description, or any sheafification of the precosheaf).*

Proof. Functoriality of $O \mapsto \mathcal{A}(O)$ is isotony together with $\iota_{O'O''} \circ \iota_{OO'} = \iota_{OO''}$ for $O \subseteq O' \subseteq O''$, and $\iota_{OO} = \text{id}$; the arrows raise the region, so the functor is covariant, i.e. a functor on Reg rather than on Reg^{op} . A presheaf on $(\text{Reg}, \subseteq)$ is by definition a functor on Reg^{op} , whose structure maps lower the region; these are the opposite of the $\iota_{OO'}$ and are not produced by isotony. The colimit gluing is the standard inductive-limit (quasilocal) construction, which is the cosheaf condition; a sheaf with restrictions is obtained only after a localization $\rho_{O'O}$ is fixed. $\square$

We deliberately do not impose the full Haag–Kastler AQFT package (microcausality, covariance, a vacuum); only the algebraic carrier role is kept: local regions/patches carry observable algebras. The minimal load-bearing content of a patch U_i is its local observable algebra

$$\mathcal{A}_i := \mathcal{A}(U_i), \quad (2)$$

a unital C^* -algebra. Occupied subspaces, Fermi projections, states, measurement effects and local unitaries are not primitive: they are selected inside $\mathcal{A}_i$ only after a realization/readout has been chosen (§2).

Remark 1.2 (Sector selection is downstream). The primitive fibre is the full observable algebra $\mathcal{A}_i$, not an occupied subspace. The Chern-sector reduction is downstream: choose a local Hamiltonian $H_i \in \mathcal{A}_i$, form the Fermi projection $P_i = \chi_{(-\infty, E_F)}(H_i) \in \pi_i(\mathcal{A}_i)''$ and the occupied subspace $E_i = \text{Ran } P_i$. The general relational carrier $\mathcal{A}_i$ is thereby separated from the specific Chern readout:

$$\text{local observable algebra} \rightsquigarrow \text{chosen projection} \rightsquigarrow \text{Chern/NCG curvature.}$$

1.2 Stage 1.5: the strict relational descent and the sheaf

On a relational cover $\{U_i\}$ the sheaf is built not from whole-algebra identifications but from genuine descent data. To each patch we attach a local algebra-sheaf $\mathcal{A}_i$ on U_i . By Lemma 1.1 the bare assignment $V \mapsto \mathcal{A}(V)$ for $V \subseteq U_i$ is a precosheaf, not a presheaf; we therefore fix once and for all the localization $\rho_{V'V} : \mathcal{A}(V') \rightarrow \mathcal{A}(V)$ of Lemma 1.1 and define $\mathcal{A}_i$ as the sheaf obtained by sheaffying the resulting presheaf $(V \mapsto \mathcal{A}(V), \rho_{V'V})$ on U_i . The restriction/localization maps are thus part of the data of the substrate, supplied by the chosen sheafification procedure, and are not read off isotony. The relational structure is then the gluing of these per-patch sheaves by comparison $*$ -isomorphisms defined on the overlaps,

$$\alpha_{ij} : \mathcal{A}_i|_{U_{ij}} \xrightarrow{\sim} \mathcal{A}_j|_{U_{ij}}, \quad (3)$$

satisfying the strict cocycle condition on triple overlaps,

$$\alpha_{ij}\alpha_{jk} = \alpha_{ik} \quad \text{on } U_{ijk}. \quad (4)$$

This is exactly descent data for a sheaf of observable algebras over the comparison site. When the cocycle holds strictly the data descend to a single global sheaf; the relational sheaf is the resulting local system of observable algebras,

$$\mathcal{F}_{\text{rel}} : U_i \mapsto \mathcal{A}_i, \quad \text{glued by } \alpha_{ij}. \quad (5)$$

It is relational before it is spectral, and flat: the comparisons compose on the nose, so $\mathcal{F}_{\text{rel}}$ carries no holonomy. Whatever topology the system has is, at this stage, the combinatorial topology of the cover/site — coordinate-free, periodicity-free, and agnostic to which invariant one will later compute.

Remark 1.3 (The groupoid is a presentation, not a foundation). The comparison site is the nerve $K = N(\mathcal{R}_Q)$ of the relational groupoid $\mathcal{R}_Q \rightrightarrows I$; but $\mathcal{R}_Q$ now enters only as a convenient presentation of the cover and its overlaps, not as the primitive object. The descent data $(\mathcal{A}_i, \alpha_{ij})$ are foundational; the groupoid is bookkeeping. This replaces “groupoid of contexts” by “strict descent over a comparison site.”

Remark 1.4 (The defect is already in the descent site). At the $+\pi/3$ disclination the pentagon’s isotropy carries five composable comparisons, not six, so the cover around the defect — and hence $K = N(\mathcal{R}_Q)$ — is structurally altered there before any operator is assigned. The relational topology sees the defect; the NCG realization will only measure it.

2 Stage 2: the NCG realization, the universal calculus, and the characteristic map

Only after the relational sheaf exists does one choose a realization. This is the one and only place noncommutative geometry enters — and it enters lightly: as a representation that lets the algebra be differentiated and integrated, not as the injection of a metric operator.

The functor. A realization is a functor $\pi_{\text{NCG}} : \mathcal{R}_Q \rightarrow \text{Hilb}_{\mathcal{A}}$ that represents the local algebras and implements the comparisons,

$$\mathcal{A}_i \mapsto (\pi_i : \mathcal{A}_i \rightarrow B(H_i)), \quad \alpha_{ij} \mapsto V_{ij}, \quad \mathcal{R}_Q \mapsto \mathcal{A}_G, \quad (6)$$

with V_{ij} the implementer of the comparison in the representation, $V_{ij}\pi_i(a)V_{ij}^* = \pi_j(\alpha_{ij}(a))$, and $\mathcal{A}_G$ the groupoid (Bellissard observable) algebra on $H = \bigoplus_i H_i$ with faithful trace T . The occupied sector is selected inside the represented algebra, $P_i \in \pi_i(\mathcal{A}_i)''$, $E_i = \text{Ran } P_i$. The realized sheaf is the compressed local calculus $U_i \mapsto P_i \Omega_u^\bullet(\mathcal{A}_G) P_i$; restriction is compression by sub-projectors, gluing is by the implementers.

Definition 2.1 (Functorial implementer). V_{ij} is the operator that implements the strict descent isomorphism α_{ij} on the represented algebra,

$$V_{ij}\pi_i(a)V_{ij}^* = \pi_j(\alpha_{ij}(a)) \quad (a \in \mathcal{A}_i),$$

determined by α_{ij} up to the unitary group of the commutant $\pi_j(\mathcal{A}_j)'$. It is a representation-theoretic datum attached to the algebraic comparison α_{ij} ; it is not the parallel transport of any occupied bundle, and its definition uses no Hamiltonian and no Fermi projection.

Proposition 2.2 (Implementers are not Berry/Kato transports). *Let (V_{ij}) be functorial implementers of a strict cocycle (α_{ij}) realized by the functor π_{NCG} , and — after a sector P_i has been chosen — let T_{ij} denote the Berry/Kato adiabatic parallel transport of the occupied subbundle $E_i = \text{Ran } P_i$. Then*

- (i) the implementer cocycle is flat: $V_{ij}V_{jk} = V_{ik}$ strictly, hence the triangular holonomy $V_{ki}V_{jk}V_{ij} = 1$;
- (ii) the physical transport is, in general, not flat: its triangular holonomy $T_{ki}T_{jk}T_{ij} \neq 1$, and the obstruction to trivializing it is the first Chern class of the occupied bundle;
- (iii) consequently $V_{ij} \neq T_{ij}$ in general. The Wilson/Kato links of the physical occupied bundle appear only downstream of sector selection, as the determinant phases on E_i ,

$$g_{ij} = \frac{\det(V_{ij} \upharpoonright_{E_i})}{|\det(V_{ij} \upharpoonright_{E_i})|},$$

whose holonomy is the realized curvature/defect class of §§3–7, not a datum of the Stage-1 descent cocycle.

Proof. (i) is functoriality of π_{NCG} applied to $\alpha_{ij}\alpha_{jk} = \alpha_{ik}$ and $\alpha_{ii} = \text{id}$. (ii) is the standard Berry/Wilson statement: adiabatic transport of a rank- r bundle over a triangulated base composes up to a $U(r)$ holonomy whose normalized trace over a 2-cycle integrates the Berry curvature, the integral obstruction being c_1 ; flatness would force $c_1 = 0$. (iii) follows: were $V_{ij} = T_{ij}$, (i) and (ii) would contradict whenever $c_1 \neq 0$. The determinant phase g_{ij} is the compression of the implementer to the occupied subspace, manifestly post- P_i , and is the discrete Wilson/Berry link whose product is read out in §6. $\square$

The differential is universal; the geometry is in the characteristic map. The differential of the calculus is not chosen by the realization. It is the *universal* differential calculus $\Omega_u^\bullet(\mathcal{A}_G)$, $da = 1 \otimes a - a \otimes 1$ extended by the graded Leibniz rule: canonical, purely algebraic, and functorial. Functoriality is the point — every comparison α_{ij} lifts canonically to $\Omega_u(\alpha_{ij})$ intertwining d , and the strict cocycle lifts strictly, so the universal calculus is carried by the substrate, glued for free, and stays flat. It is also featureless: it injects no metric and no curvature of its own. Of everything the realization supplies, the one piece of analytic content needed downstream is the faithful trace T on $\mathcal{A}_G$, which makes the integration converge; the invariant-selecting geometry — the position derivations $\partial_j = [X_j, \cdot]$ — lives not in the differential but in the cyclic characteristic map τ of §4. Hence the thesis of the paper: the noncommutative content of the theory is split between a free, universal differential and a chosen characteristic map; supply the position cocycle that defines τ , fix the trace, and the rest of the machinery turns automatically.

Remark 2.3 (Functoriality keeps the substrate flat). Because π_{NCG} preserves composition, $V_{ij}V_{jk} = V_{ik}$ holds strictly, so the triangular holonomy $V_{ki}V_{jk}V_{ij} = 1$ is trivial: a strict realization leaves no holonomy-curvature, and the universal calculus glues flatly. The universal curvature $F = P(dP)^2$ is therefore featureless until paired with τ ; all invariant content comes from the characteristic map. This is the point, not a defect: it commits the construction to the Chern–Weil picture and, as §3 shows, it is exactly what protects the free Bianchi identity. On a defect, the position cocycle that defines τ cannot use global coordinates — see §8.

3 Connection, curvature, and the automatic Bianchi identity

Curvature is born here, when d first acts on the selected projection. The connection is the universal derivative of the implementer and the curvature is its compressed square:

$$a_{ij} := dV_{ij}, \quad F = P(dP)^2, \quad F_{ijk}^{\text{rel}} = P_i a_{ij} a_{jk}. \quad (7)$$

Explicitly $F_{ijk}^{\text{rel}} = P_i dV_{ij} dV_{jk}$.

Proposition 3.1 (Bianchi is automatic). *The curvature (7) is closed without any associativity hypothesis.*

- (i) *From $P^2 = P$ one gets $P(dP)P = 0$, and with Leibniz this is the Bianchi identity $\nabla F = 0$ for the Grassmann connection $\nabla = P d$.*
- (ii) *From the strictly preserved composition $V_{ij}V_{jk} = V_{ik}$ and Leibniz,*

$$dV_{ik} = dV_{ij} V_{jk} + V_{ij} dV_{jk},$$

the Maurer–Cartan relation for a_{ij} , whence $\check{d}F_{ijk}^{\text{rel}} = 0$.

Proof. Differentiating $P^2 = P$ gives $(dP)P + P(dP) = dP$; compressing by P yields $P(dP)P = 0$, and the standard projector computation gives $\nabla F = 0$. Part (ii) is Leibniz applied to the functorial composition law of the implementers. $\square$

Closure is a calculus identity, and Proposition 3.1(ii) makes explicit that it is functoriality of π_{NCG} — preservation of $V_{ij}V_{jk} = V_{ik}$ — that carries it. No twisted algebra and no associativity theorem are needed.

4 The characteristic map and the relational Chern class

Closure of the universal curvature does not by itself yield a scalar; the bridge is a cyclic cocycle. This is the single, proper role of the characteristic map, and it is where the invariant is both selected and integrated.

Definition 4.1 (Characteristic map). For a cyclic cocycle τ on $\mathcal{A}_G$ ($b\tau = 0$),

$$\chi_\tau : F_{ijk}^{\text{rel}} \longmapsto c_{ijk} := \tau(F_{ijk}^{\text{rel}}) = \tau(P_i dV_{ij} dV_{jk}) \in \mathbb{C}. \quad (8)$$

For the Chern/Hall family τ is the Connes–Chern (Bellissard area) cocycle, of density $\frac{1}{2\pi i}\tau(P[\partial_1 P, \partial_2 P])$, $\partial_j = [X_j, \cdot]$. The position derivations ∂_j reside here, in τ — not in the differential — so the characteristic map is where the invariant-selecting geometry of the realization is carried.

Theorem 4.2 (Čech–Connes class and its integral lift). *If τ is a cyclic cocycle and F^{rel} is Bianchi-closed (Proposition 3.1), then $c = \chi_\tau(F^{\text{rel}})$ is a Čech 2-cocycle and defines $[c] \in \check{H}^2(N(\mathcal{R}_Q); \mathbb{C})$ in the total cohomology of the Čech–Connes bicomplex (Appendix A). The complex class lifts to the integers in two complementary ways:*

- (i) (bulk, absolute) *the Connes pairing $\langle \tau, [P] \rangle \in \mathbb{Z}$ of the area cocycle with $[P] \in K_0(\mathcal{A}_G)$ is an index, integral because τ is the Connes–Chern character of a finitely-summable Fredholm module;*
- (ii) (defect, relative) *for the rank-one occupied band the determinant phases g_{ij} are the $U(1)$ transition functions of the occupied line bundle, and the exponential sequence $U(1) \rightarrow \mathbb{R} \rightarrow \mathbb{Z}$ Bocksteins them on the disk $\text{St}(d)$, where $\check{H}^2(\text{St}(d), \partial \text{St}(d)) \cong \mathbb{Z}$.*

In both forms one obtains the relational first Chern class $c_1^{\text{rel}} \in \check{H}^2(N(\mathcal{R}_Q); \mathbb{Z})$.

The integral lift is the point: a Chern class is quantized, and that is visible only over $\mathbb{Z}$. The bulk number is an index (needs the trace and the summable Fredholm structure); the defect number is a determinant winding (needs only the implementer determinants).

Remark 4.3 (The characteristic map is modular). The substrate, the realization, and Bianchi are fixed; the cyclic cocycle τ alone selects the invariant — area cocycle $\Rightarrow$ Chern/Hall, higher cocycles $\Rightarrow$ higher Chern, odd cocycles with a symmetry $\Rightarrow \mathbb{Z}_2$, the canonical trace $\Rightarrow$ gap-labeling, a quasi-energy $\tau \Rightarrow$ Floquet. An index cocycle gives a quantized integer; a non-index spectral functional gives non-quantized data (transport coefficients, resonances) on the same substrate. The differential and curvature never change; only χ_τ does.

5 The crystalline equivalence (foundational hypothesis)

The validation against known topology — that in the translation-invariant sector the relational class is the Bloch–Berry Chern class — remains unproved and is recorded as the foundational hypothesis.

Hypothesis 5.1 (Crystalline equivalence). *For the periodic Haldane model there is a translation-invariant sector $\check{H}^2(N(\mathcal{R}_Q); \mathbb{Z})_{\text{cryst}} \subseteq \check{H}^2(N(\mathcal{R}_Q); \mathbb{Z})$ and a comparison $\Theta : \check{H}^2(N(\mathcal{R}_Q); \mathbb{Z})_{\text{cryst}} \rightarrow H^2(T^2; \mathbb{Z})$ with $\Theta(c_1^{\text{rel}}) = c_1(B)$, $B \rightarrow T^2$ the Bloch bundle, carrying the Connes–Chern density to the Berry curvature.*

Corollary 5.2. *Assume Hypothesis 5.1. Then $\langle \Theta(c_1^{\text{rel}}), [T^2] \rangle = \langle c_1(B), [T^2] \rangle = C$, the Haldane Chern number.*

This is the external bridge to Bloch theory; it requires periodicity (a torus). The disclination class lies outside the crystalline sector and is reached without it.

6 The disclination and the relative class

Let d be a pentagonal disclination, $\text{St}(d) \subset N(\mathcal{R}_Q)$ its star, $\partial \text{St}(d)$ its boundary. With c^{ref} the cochain of the defect-free reference, the relative cocycle $\rho_d = c - c^{\text{ref}}$ is supported on $\text{St}(d)$ when defect and reference agree outside it, and

$$c_d^{\text{rel}} := [\rho_d] \in \check{H}^2(\text{St}(d), \partial \text{St}(d)), \quad \Phi_d = \prod_{(ij) \in L_d} g_{ij}, \quad g_{ij} = \frac{\det(V_{ij} \upharpoonright_{E_i})}{|\det(V_{ij} \upharpoonright_{E_i})|}, \quad (9)$$

the pairing of c_d^{rel} with the loop L_d around the pentagon (Figure 1). Here, by Proposition 2.2(iii), the g_{ij} are the Wilson/Berry determinant phases of the implementers compressed to the occupied subbundle E_i , hence post-sector-selection readouts rather than Stage-1 data.

The next proposition isolates why a flat substrate can nonetheless carry a nonzero Φ_d : strict composability is a bulk statement, and the disclination is a controlled, localized violation of it.

Proposition 6.1 (Bulk strictness vs. defect holonomy). *Suppose the strict cocycle $\alpha_{ij}\alpha_{jk} = \alpha_{ik}$ holds on every bulk triple overlap, so that $V_{ij}V_{jk} = V_{ik}$ there. Then:*

- (i) *around any bulk loop the implementers telescope and $\Phi = 1$ identically;*
- (ii) *a nonzero pentagon holonomy $\Phi_d \neq \Phi_d^{\text{ref}}$ requires the loop L_d to fail strict closure — which is exactly the structural alteration of the descent site at d (five composable comparisons, not six), the disclination conjugating the incident links by the Frank-wedge rotation $R(\pi/3)$;*
- (iii) *hence Φ_d measures the localized failure of strict closure at d ; it is a transport-determinant (post- P_i , independent of τ) quantity, consistent with the substrate being flat in the bulk.*

Proof. (i) Multiplicativity of $\det$ on E_i and strict composition give $\prod_{(ij) \in L} g_{ij} = \det(\prod_L V_{ij}) \upharpoonright_E / |\cdot|$; a closed bulk loop composes to the identity, so the determinant phase is 1. (ii) If $V_{ij}V_{jk} = V_{ik}$ held around L_d as well, the same telescoping would force $\Phi_d = 1 = \Phi_d^{\text{ref}}$; a nonzero relative phase therefore obstructs strict closure on L_d , which is the five-fold coordination of Remark 1.4 and the $R(\pi/3)$ conjugation. (iii) is the combination, with the characteristic map’s derivations entering nowhere in g_{ij} . $\square$

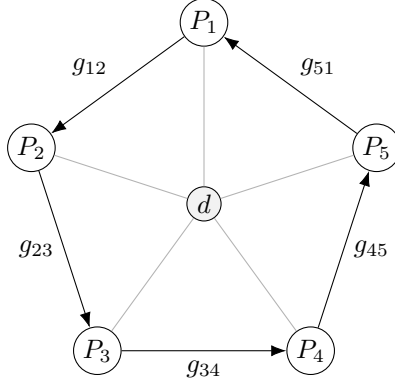


Figure 1: The star $\text{St}(d)$: five contexts, five implementers, five triangles $\{d, i, i+1\}$ carrying $F_{d,i,i+1}^{\text{rel}}$. Curvature is supported here; c_1^{rel} localizes as c_d^{rel} , paired with L_d as $\Phi_d = \prod_i g_{i,i+1}$.

7 The Local Defect Theorem

Theorem 7.1 (Local Defect Theorem). *Assume bulk flatness: $F = F^{\text{ref}}$ and $P = P^{\text{ref}}$ off $\text{St}(d)$. Then c_d^{rel} is gauge- and Morita-invariant, supported on $\text{St}(d)$, and*

$$[c_d] \neq 0 \iff c_d^{\text{rel}} \neq 0 \iff \Phi_d \neq \Phi_d^{\text{ref}}.$$

Moreover the deciding quantity Φ_d is built from transport determinants g_{ij} alone, so for the determinant (Chern) characteristic map the defect decision is independent of the characteristic map's derivations.

Sketch. Invariance: $[c]$ is a Čech–Connes class; a frame change shifts a_{ij} by an inner term traced into a coboundary by cyclicity of τ , and a realization change presenting the same $\mathcal{R}_Q$ is Morita. Localization: bulk flatness makes ρ_d vanish off $\text{St}(d)$; the determinant phases telescope along interior transports, leaving the boundary ratio $\Phi_d/\Phi_d^{\text{ref}}$. The last clause follows because $g_{ij} = \det(V_{ij} \upharpoonright_{E_i})/|\det(V_{ij} \upharpoonright_{E_i})|$ does not involve the position derivations. $\square$

So the defect monodromy is a transport-level quantity (post-functor, post- P_i , independent of τ), while the bulk Chern class is an index/characteristic-map-level quantity; this theorem is the bridge that lets the former decide the latter.

8 Two open constructions

The two-stage formulation concentrates all genuine difficulty in two places.

Open construction 8.1 (The position cocycle and the characteristic map). On a disclination there is no single-valued position, so the derivations cannot be the flat-plane ∂_X, ∂_Y . They must be built from a position 1-cocycle on the groupoid, $X : \mathcal{R}_Q \rightarrow \mathbb{R}^2$ with $X(\gamma\gamma') = X(\gamma) + \gamma \cdot X(\gamma')$, giving relational derivations $\partial_j = [X_j, \cdot]$ without global coordinates. These derivations, together with the trace T , define the area cocycle τ — the characteristic map — and so, by the thesis of §2, carry the entire invariant-selecting geometry. What the construction pivots on is the existence of the position cocycle and the finiteness/summability of T that makes $\langle \tau, [P] \rangle$ land in $\mathbb{Z}$; no self-adjoint Dirac operator and no global coordinates are required.

Open construction 8.2 (The realization comparison). π_{NCG} induces a map on cohomology; one must show it is an isomorphism on the sector of interest, so that the relational Čech class and the realized NCG class agree. Equivalently, the bulk Bellissard curvature $P[\partial_1 P, \partial_2 P]$ and the simplicial F_{ijk}^{rel} represent the same Čech–Connes class. This comparison is the one nontrivial internal identity; it is periodicity-free, and distinct from Hypothesis 5.1 (which is the external bridge to Bloch theory).

9 To-do list: computing the Haldane disclination

For the Haldane model in the topological regime ($C = \pm 1$) the pipeline is finite.

Stage 1 — local algebra substrate.

- Attach local observable algebras $\mathcal{A}_i$ around the defect. Record the comparison $*$ -isomorphisms α_{ij} on overlaps and the descent site $K = N(\mathcal{R}_Q)$, including the five-fold star at d . No sector is selected at this stage: H_i, P_i, E_i are downstream (§2).

Stage 2 — realize, select sector, and differentiate.

- Apply π_{NCG} : $\pi_i : \mathcal{A}_i \rightarrow B(H_i)$, $\alpha_{ij} \mapsto V_{ij}$ (functorial implementers from the descent isomorphisms, not Berry/Kato transports; Definition 2.1, Proposition 2.2), $\mathcal{R}_Q \mapsto \mathcal{A}_G$ with faithful trace T .
- After realization/readout, select $H_i \in \mathcal{A}_i$, form the Fermi projection $P_i = \chi_{(-\infty, E_F)}(H_i) \in \pi_i(\mathcal{A}_i)''$, and set $E_i = \text{Ran } P_i$ (frames e_i if used).
- Take the universal differential d (free, functorial); build the position cocycle X and derivations $\partial_j = [X_j, \cdot]$ defining τ (Open construction 8.1); form $a_{ij} = dV_{ij}$ and $F_{ijk}^{\text{rel}} = P_i a_{ij} a_{jk}$.
- Closure is structural: verify $dV_{ik} = dV_{ij} V_{jk} + V_{ij} dV_{jk}$ (Proposition 3.1); no associativity check.

Stage 3 — characteristic map and class.

- Apply the Connes–Chern area cocycle: $c_{ijk} = \tau(F_{ijk}^{\text{rel}})$; integralize the bulk number as the index pairing $\langle \tau, [P] \rangle$ to get $c_1^{\text{rel}} = [c] \in \check{H}^2(N(\mathcal{R}_Q); \mathbb{Z})$.
- Cross-check $\langle c_1^{\text{rel}}, [\Sigma] \rangle$ against the bulk Bellissard number $\frac{1}{2\pi i} \tau(P[\partial_1 P, \partial_2 P])$.

Stage 4 — localize and decide (transport-level).

- On $\text{St}(d)$ form $\rho_d = c - c^{\text{ref}}$ (nonzero by five- vs. six-fold coordination; Proposition 6.1); $\text{St}(d)$ is a disk so $\check{H}^2(\text{St}(d), \partial \text{St}(d)) \cong \mathbb{Z}$.
- Compute the links $g_{i,i+1} = \det(V_{i,i+1} \upharpoonright_{E_i}) / |\det(V_{i,i+1} \upharpoonright_{E_i})|$ and $\Phi_d = \prod_i g_{i,i+1}$, Φ_d^{ref} . The disclination conjugates the incident $V_{i,i+1}$ by $R(\pi/3)$ (the Frank wedge).
- Decide: $\Delta\Phi = \arg(\Phi_d / \Phi_d^{\text{ref}}) \in \mathbb{R}/2\pi\mathbb{Z}$, with $c_d^{\text{rel}} \neq 0 \iff \Delta\Phi \notin 2\pi\mathbb{Z}$. By Theorem 7.1 this step needs only the transports, not the characteristic map's derivations; the expected form is $\Delta\Phi \propto (\text{Frank angle}/2\pi)$ weighted by bulk band data.

10 Status and outlook

Relations come first and are flat; the realization functor represents the algebra and supplies a trace, the differential is the universal calculus carried for free by the substrate, and curvature is born only when d acts on the selected projection. The geometry that distinguishes invariants enters at one point — the cyclic characteristic map, whose position derivations live in τ , not in any Dirac operator. Closure is then a calculus identity that functoriality protects (Proposition 3.1), and a single cyclic cocycle bridges the universal curvature to a Čech–Connes class and selects the invariant. The crystalline equivalence (Hypothesis 5.1) remains the open external validation against Bloch theory and is not assumed. All genuine difficulty is concentrated in the position cocycle defining the characteristic map (Open construction 8.1) and the realization comparison (Open construction 8.2); the disclination decision, by contrast, reduces to a finite transport-determinant holonomy.

A The Čech–Connes bicomplex

The cover direction supplies the Čech differential $\check{\delta}$ on $N(\mathcal{R}_Q)$; the operator direction supplies the cyclic (b, B) differentials of $\Omega_u^\bullet(\mathcal{A}_G)$. Curvature is a total representative: F^{rel} is $\check{\delta}$ -closed by Bianchi (Proposition 3.1) and τ is b -closed; χ_τ contracts the operator legs, leaving a $\check{\delta}$ -cocycle of class $[c] \in \check{H}^2(N(\mathcal{R}_Q); \mathbb{C})$, integralized to c_1^{rel} by the index pairing (bulk) or the exponential sequence (defect). This is the standard setting for a relational NCG Chern–Weil statement: noncommutative curvature, abelian cohomology, mediated by a cyclic characteristic map [1, 2, 5].

B Dictionary: relational $\rightarrow$ realized $\rightarrow$ Haldane

Relational (Stage 1 / 1.5)	NCG realization (Stage 2)	Haldane
local algebra $\mathcal{A}_i$	represented $\pi_i(\mathcal{A}_i) \subset B(H_i)$	local observable algebra
selected sector $P_i \in \pi_i(\mathcal{A}_i)''$	$E_i = \text{Ran } P_i$	Fermi projection
comparison α_{ij} (strict $*$ -iso)	implementer V_{ij} , $V_{ij}V_{jk} = V_{ik}$	$\chi_{(-\infty, E_F)}(H_i)$ strict implementer (not Berry/Kato transport)
descent site $K = N(\mathcal{R}_Q)$	algebra $\mathcal{A}_G$, trace T , universal Ω_u	observables, position cocycle
sheaf $\mathcal{F}_{\text{rel}}(U_i) = \mathcal{A}_i$ (per-patch, glued by α_{ij})	$P_i \Omega_u^\bullet(\mathcal{A}_G) P_i$	local calculi
(flat: strict descent)	connection $a_{ij} = dV_{ij}$	transport derivative
—	curvature $F = P(dP)^2$	$\pi/3$ at the pentagon
combinatorial closure	$P^2 = P$ & Leibniz $\Rightarrow \check{\delta}F = 0$	automatic Bianchi
—	χ_τ (Connes–Chern), $\partial_j = [X_j, \cdot]$	area cocycle
relational class $[c] \in \check{H}^2$; lift $c_1^{\text{rel}} \in \check{H}^2(\mathbb{Z})$	$\langle \tau, [P] \rangle$ index pairing; defect c_d^{rel}	$\Phi_d = \prod g_{ij}$, Wilson dets on E_i (post- P_i)

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